

PROJECTEXPO 2026

SMART WILD PIG DETECTION AND ALERT SYTEM USING IoT AND AI

Phase1- Deliverables

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INTRODUCTION:

Agriculture is one of the most important sectors that supports food production and the economy. However, farmers often face significant losses due to wild animals entering agricultural fields and damaging crops. Among these animals, wild pigs are one of the major causes of crop destruction because they feed on crops and trample fields, leading to reduced yield and financial loss.

Traditional methods such as manual guarding and fencing require continuous human effort, are time-consuming, and may not always be effective. With the advancement of the Internet of Things (IoT), it is possible to develop smart systems that automatically monitor agricultural fields and provide instant alerts when wild pigs are detected.

This project, "Smart Wild Pig Detection and Alert System Using IoT and Sensors," uses a PIR motion sensor and an ultrasonic sensor to detect wild pig movement near the farm. An ESP32 microcontroller processes the sensor data and activates a buzzer to alert nearby people. At the same time, the system sends a real-time notification to the farmer through an IoT platform, allowing quick action to prevent crop damage.

The proposed system is low-cost, energy-efficient, and easy to install, making it suitable for small and medium-sized farms. It helps farmers monitor their fields continuously without requiring constant human supervision. This project aims to improve crop protection and reduce agricultural losses through smart sensor technology and IoT-based communication.

Problem Statement:

Wild pigs frequently enter agricultural fields and cause severe crop damage, resulting in significant economic losses for farmers. Traditional methods such as manual guarding are time-consuming, labor-intensive, and cannot provide continuous monitoring. There is a need for an automated and affordable system that can detect wild pig movement and immediately notify farmers. This project proposes a Smart Wild Pig Detection System using IoT and Sensors. The system uses PIR and ultrasonic sensors to detect movement and distance. When a wild pig is detected, an alarm (buzzer) is activated, and an IoT notification is sent to the farmer. The system provides continuous monitoring, reduces crop damage, and improves farm security with minimal human intervention.

Abstract :

Wild pig intrusion is a major problem in agricultural fields, causing extensive crop damage and financial losses to farmers. Conventional methods such as manual guarding require continuous human effort and are often ineffective for protecting large agricultural areas. To overcome these limitations, this project presents a Smart Wild Pig Detection System using IoT and Sensors.

The proposed system consists of a PIR motion sensor, an ultrasonic sensor, an ESP32 microcontroller, a buzzer, and an IoT platform. The PIR sensor detects movement, while the ultrasonic sensor measures the distance of the approaching object to improve detection accuracy. The ESP32 processes the sensor data and identifies possible wild pig movement. Once the animal is detected, the system immediately activates a buzzer to alert nearby people and simultaneously sends a real-time notification to the farmer through an IoT platform or mobile application.

The proposed solution is affordable, energy-efficient, and easy to install in agricultural fields. It provides continuous monitoring without requiring constant human supervision. By combining sensor-based detection with IoT communication, the system enables farmers to respond quickly and reduce crop damage. The project offers a practical and reliable solution for protecting agricultural fields. In the future, the system can be enhanced by integrating cameras, artificial intelligence, GPS tracking, or solar power to improve its performance and coverage.

Objectives:

- ❑ To detect wild pig movement using PIR and ultrasonic sensors.
- ❑ To continuously monitor agricultural fields.
- ❑ To activate a buzzer when wild pig movement is detected.
- ❑ To send real-time notifications to farmers using IoT.
- ❑ To reduce crop damage through an automated monitoring system.
- ❑ To develop a low-cost and energy-efficient solution.

Expected Outcomes:

- ❑ Accurate detection of wild pig movement.
- ❑ Automatic activation of the buzzer upon detection.
- ❑ Real-time IoT notifications to farmers.
- ❑ Reduced crop damage and financial losses.
- ❑ Improved farm monitoring and security.

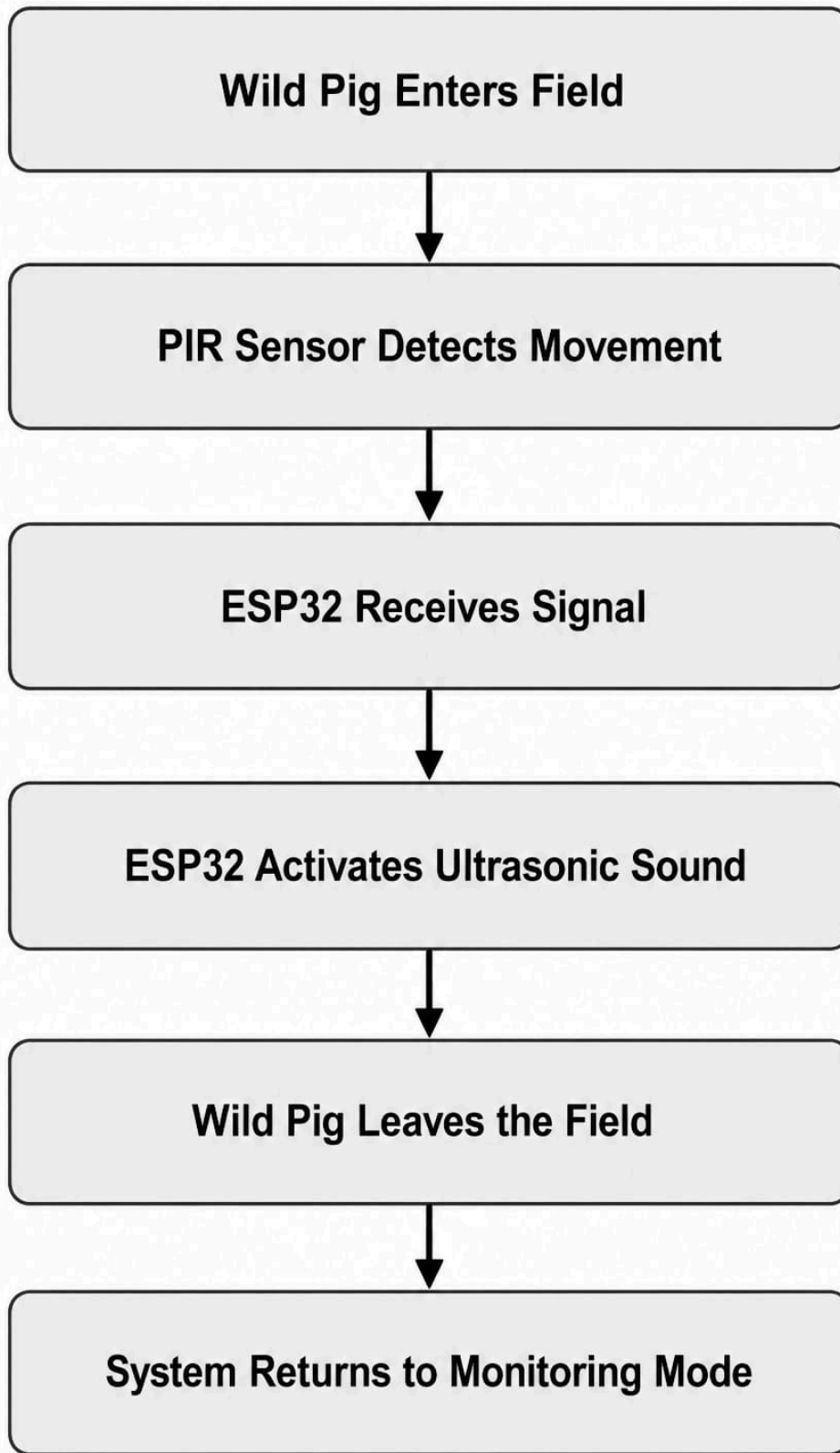
□ Low-cost, reliable, and easy-to-use system.

Literature Survey:

S.No	Author(s) & Year	Title	Technology Used	Key Findings	Limitations
1	Divya et al. (2018)	IoT-Based Wild Animal Intrusion Detection System	PIR, Camera, GSM	SMS alerts	False alarms
2	Alui& Mohamed (2023)	Animal Intrusion Monitoring	PIR, Camera, Blynk	Real-time monitoring	No species ID
3	Abraham et al. (2023)	Review of Intrusion Systems	IoT, GSM	Reviews methods	No new system
4	Miao et al. (2023)	Fog-based Agriculture	LoRa, Camera	Low latency	Complex
5	MDPI (2025)	YOLOv8 Surveillance	Raspberry Pi, YOLO	High accuracy	High processing
6	Joshi et al. (2025)	Wild Boar Detection	YOLOv8, GSM	Auto deterrent	Expensive
7	Ninan et al. (2026)	AI Wild Boar Detection	Thermal, YOLO	Instant alerts	Training needed

Existing Solutions Analysis:

S.No	Existing Solution	Technology Used	Advantages	Limitations
1	IoT-Based Wild Animal Intrusion Detection System	PIR Sensor, Camera, GSM	Sends SMS alerts and detects movement	False alarms; no species identification
2	Animal Intrusion Monitoring System	PIR Sensor, Camera, Blynk	Real-time monitoring	Cannot identify specific animal species
3	Review of Intrusion Detection Systems	IoT, GSM	Summarizes methods	No practical implementation
4	Fog-Based Agriculture Monitoring	LoRa, Camera	Low latency	Complex and costly
5	YOLOv8 Surveillance System	Raspberry Pi, YOLOv8	High accuracy	High processing requirement
6	Wild Boar Detection System	YOLOv8, GSM	Automatic detection	Expensive
7	AI-Based Wild Boar Detection	Thermal Camera, YOLO	Accurate alerts	Costly hardware



Gap Analysis:

S.No	Research Gap Identified	Proposed Solution in Our Project
1	False alarms due to motion-only detection	Use ESP32-CAM with PIR verification
2	AI systems require powerful hardware	Use low-cost ESP32 without AI
3	High implementation cost	Affordable prototype
4	Only monitoring, no deterrent	Automatic ultrasonic deterrent
5	Needs continuous internet	Standalone operation
6	High power and maintenance	Compact low-power ESP32
7	Difficult for rural deployment	Simple portable design

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